HD-021

Medium

Requested by

IT Administrator

Domain

aklab.local

Issue:

Develop PowerShell Scripts for Active Directory Administration

1. Ticket Description

The IT Administrator has requested the development of a collection of PowerShell scripts to simplify routine Active Directory administration tasks and reduce the time spent performing repetitive operations manually.

The scripts should generate useful administrative reports and automate basic employee account creation. Each script must be documented, tested in the lab environment, and produce clear output or confirmation messages.

2. Requirements

Develop the following five PowerShell scripts:

1. Active Directory User Inventory Report

- Retrieve user accounts from the aklab.local domain.
- Display employee names, usernames, departments, and account status.
- Export the results to a CSV file for administrative review.

2. Disabled Account Audit

- Identify disabled Active Directory user accounts.
- Display relevant account details, including username, department, and last-change information where available.
- Export the results to CSV for review by the administrator.

3. Security Group Membership Report

- Allow the administrator to specify an Active Directory security group.
- Retrieve and display its members.
- Export membership details, including names, usernames, and object types, to CSV.

4. Organizational Unit (OU) Inventory Report

- Retrieve all organizational units in the domain.
- Include OU names, Distinguished Names, canonical paths, and descriptions where available.
- Export the inventory to CSV for documentation and auditing.

5. Interactive Employee Account Creation

- Prompt the technician to enter the employee's first name, last name, desired username, department, target OU, and initial password.
- Verify that the username is not already in use and that the target OU exists.
- Display the entered details and request confirmation before creating the account.
- Create the account with an initial password and require a password change at the next logon.

- Display a success or error message.

Environment: DC01 (domain controller) and CLIENT01 (Windows client)

1. Issue Description

The IT Administrator has requested the development of a collection of PowerShell scripts to simplify routine Active Directory administration tasks and reduce the time spent performing repetitive operations manually.

2. Environment

- User account: AKLAB\jamie.taylor
- Shared resource: \\DC01\CompanyShare\Employees

3. Resolution

Created the following scripts:

01-Get-ADUserReport.ps1

02-Find-DisabledAccounts.ps1

03-Get-GroupMembershipReport.ps1

04-Export-OUInventory.ps1

05-New-EmployeeAccount.ps1

4. Validation

Placing the scripts inside C: drive:

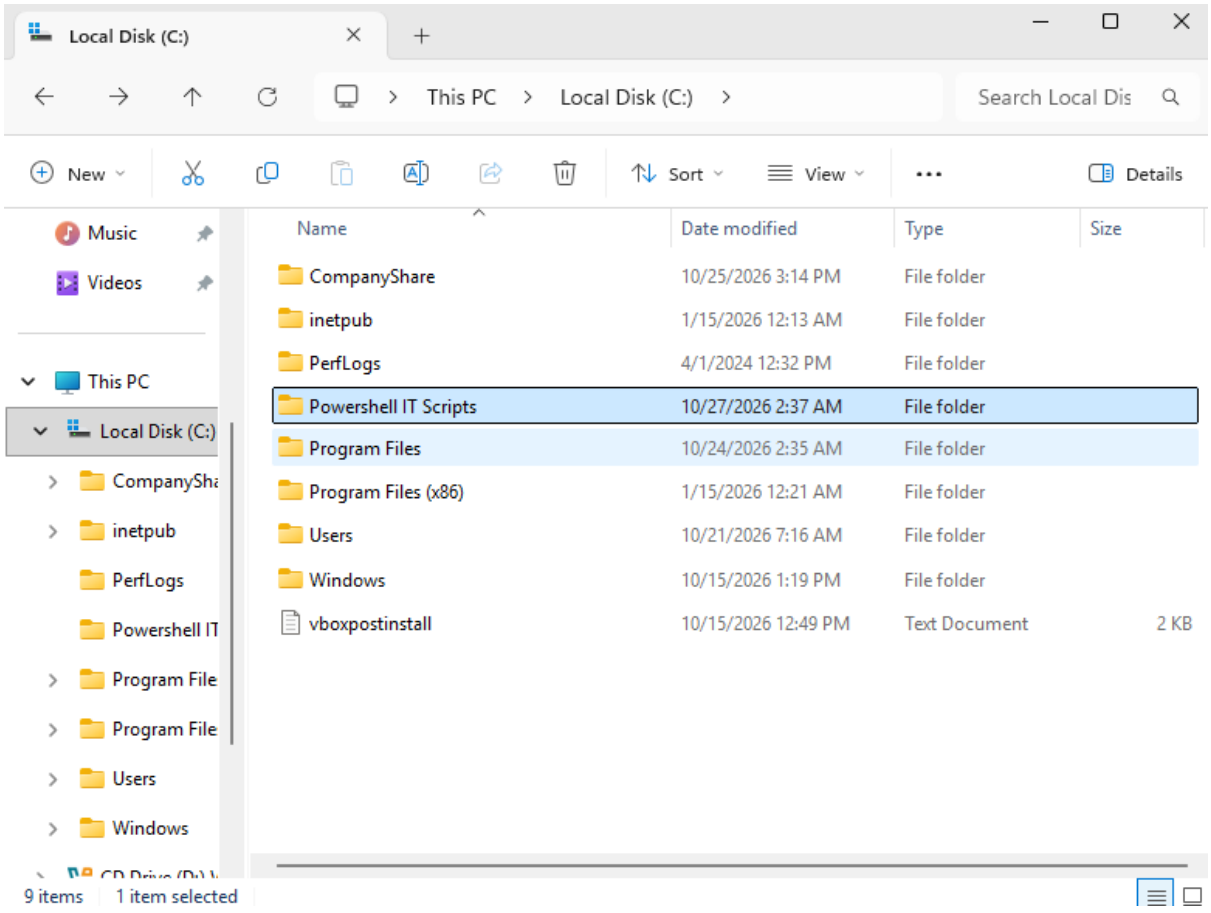


Figure 1: Creating Powershell IT Scripts folder

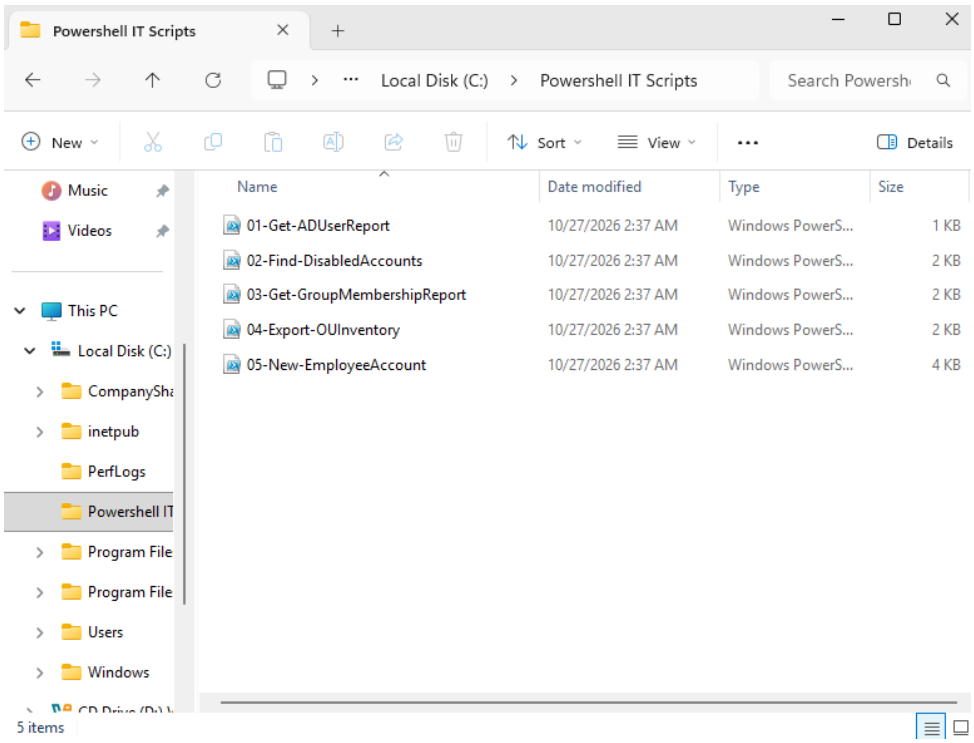


Figure 2: Pasting the five scripts inside the folder

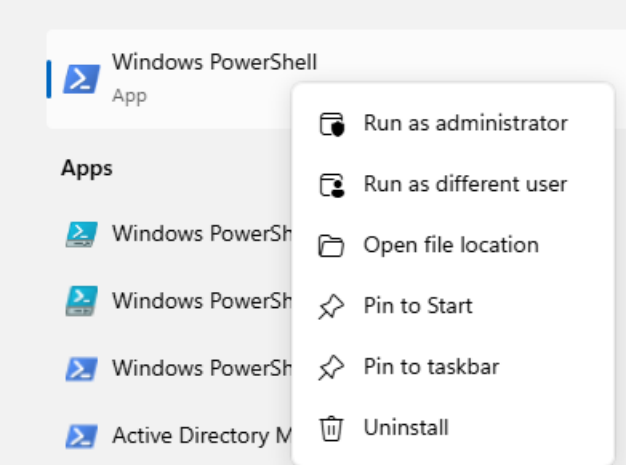


Figure 3: Running Powershell as an administrator

We can easily run a .ps1 file using these commands:

Set-Location "C:\Powershell-IT-Scripts"(Setting the location)

.\01-Get-ADUserReport.ps1 (Pasting the name of the file)

Script 1: Getting ADUser Report

```
PS C:\WINDOWS\system32> Set-Location "C:\Powershell IT Scripts"
PS C:\Powershell IT Scripts> .\01-Get-ADUserReport.ps1
User report generated successfully!
Report location: C:\Users\Administrator\Desktop\AD-User-Report.csv
Total user accounts: 13
PS C:\Powershell IT Scripts>
```

Figure 4: Script 1 result

Successfully run Script 1 and the result provided the total number of user accounts and generated a report.

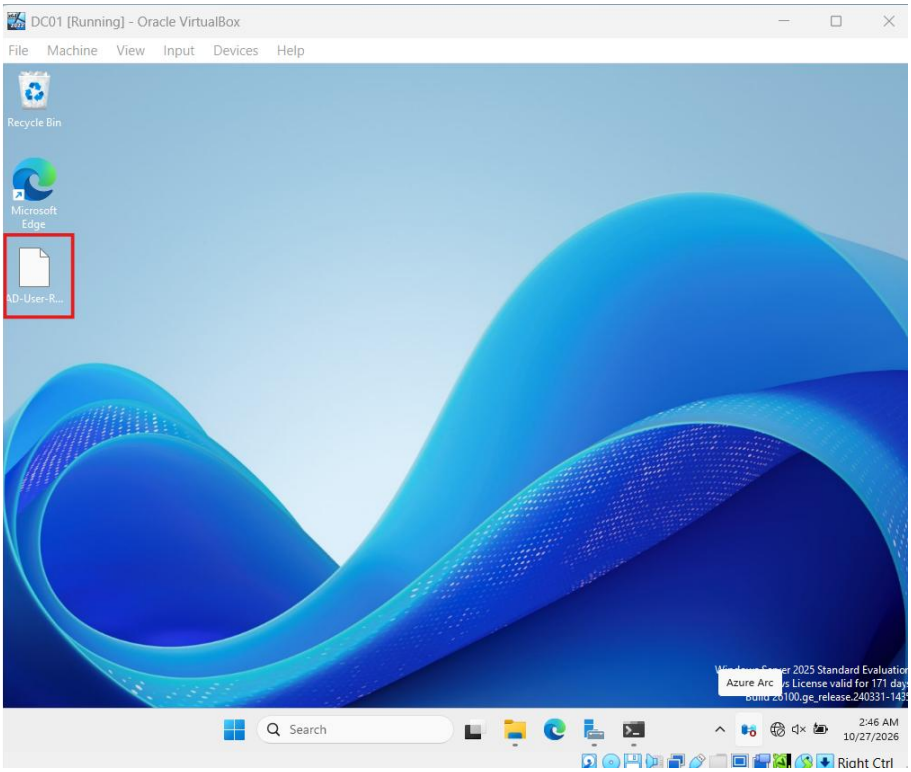


Figure 5: ADUser Report Generated

This is the file that was generated:



Figure 6: Report generated

Running Script 2:

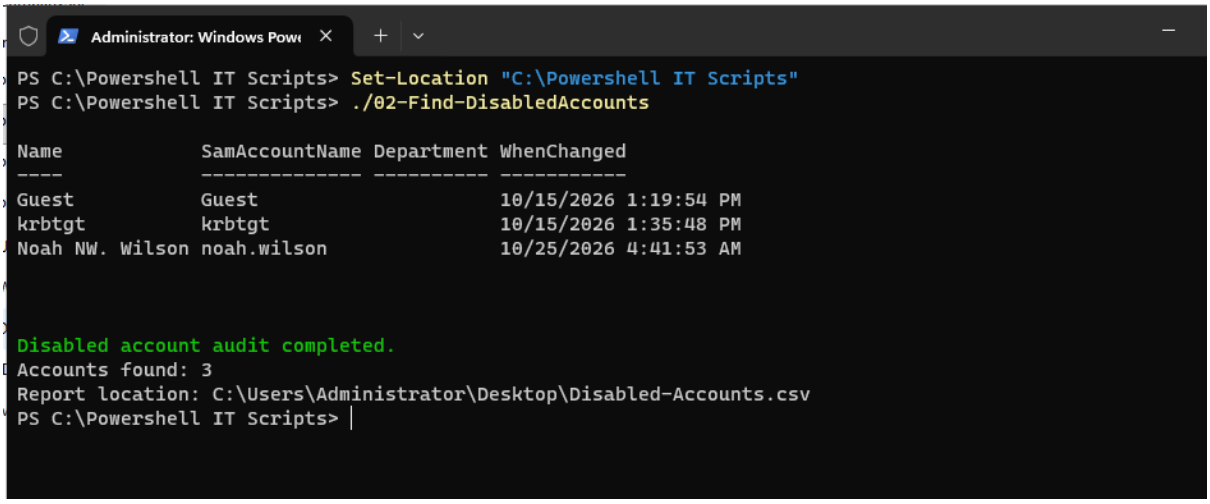


Figure 7: Script 2 run successfully

This shows that there are three disabled accounts found in the Active Directory along with their names.

Running Script 3:

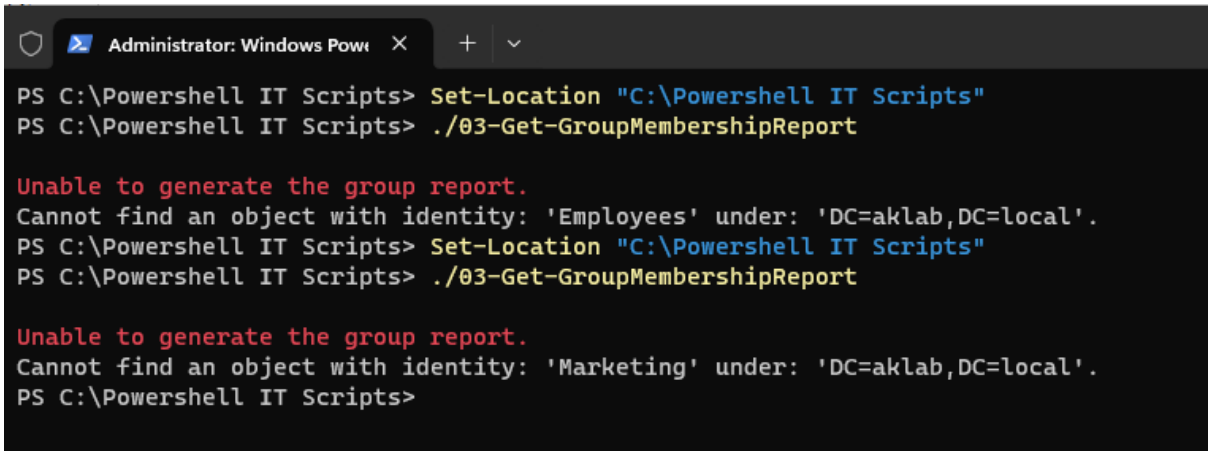


Figure 8: Running Script 3 failed

Let's try to fix the issue:

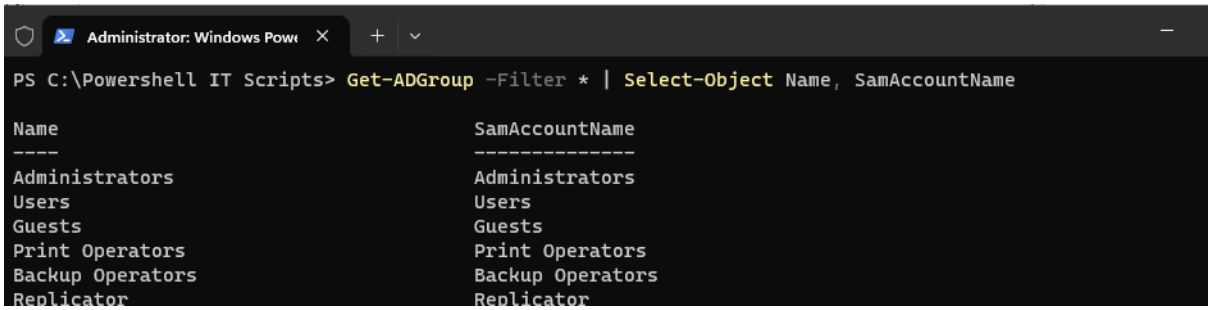


Figure 9: Troubleshooting Script 3

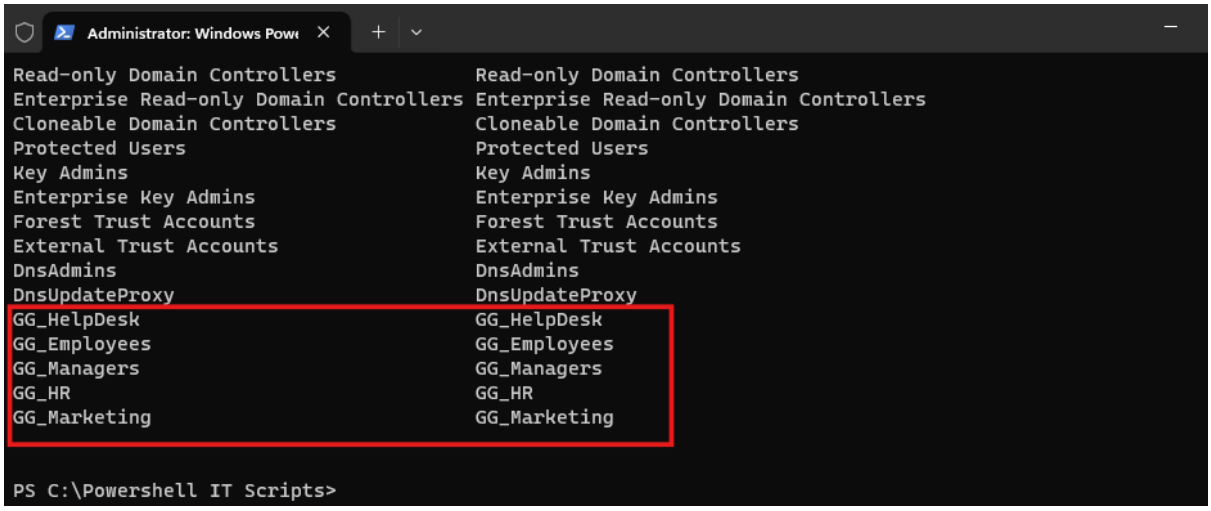


Figure 10: Troubleshooting Script 3

We can see that the groups are named as GG_Employees etc, let's try to change the name in the script and run again.

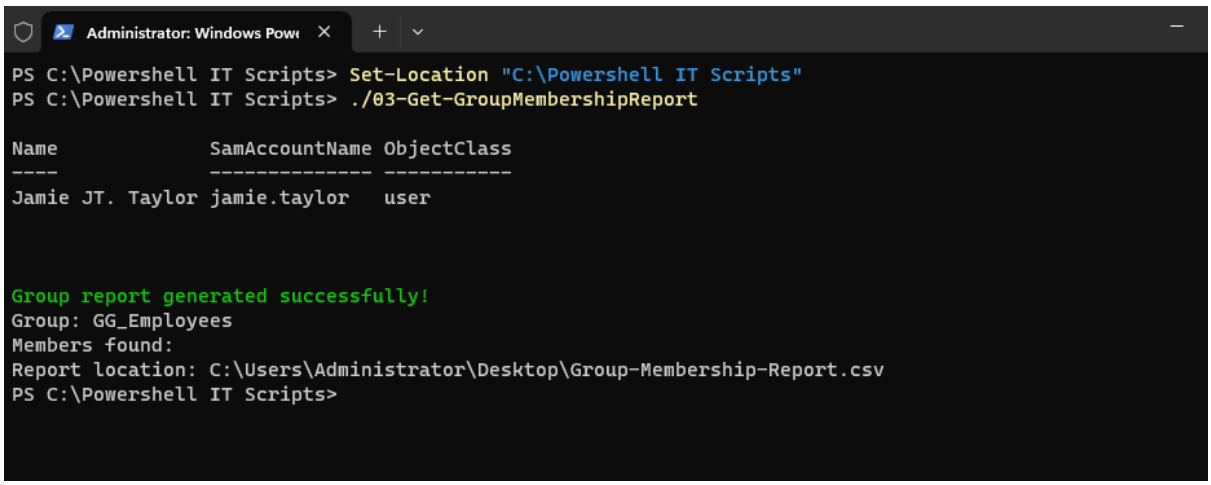


Figure 11: Troubleshooting fixed the issue

Now, its able to access the group, but let's try to edit to get the count.

After revising the code, now, we are getting the members count too:

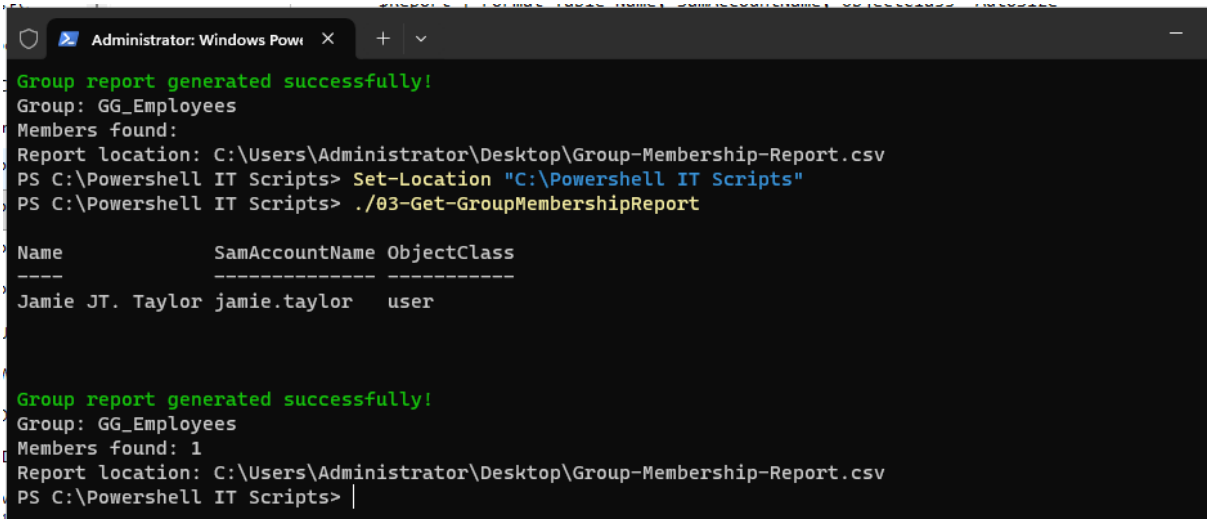


Figure 12: Script 3 runs successfully

Script 4 Testing

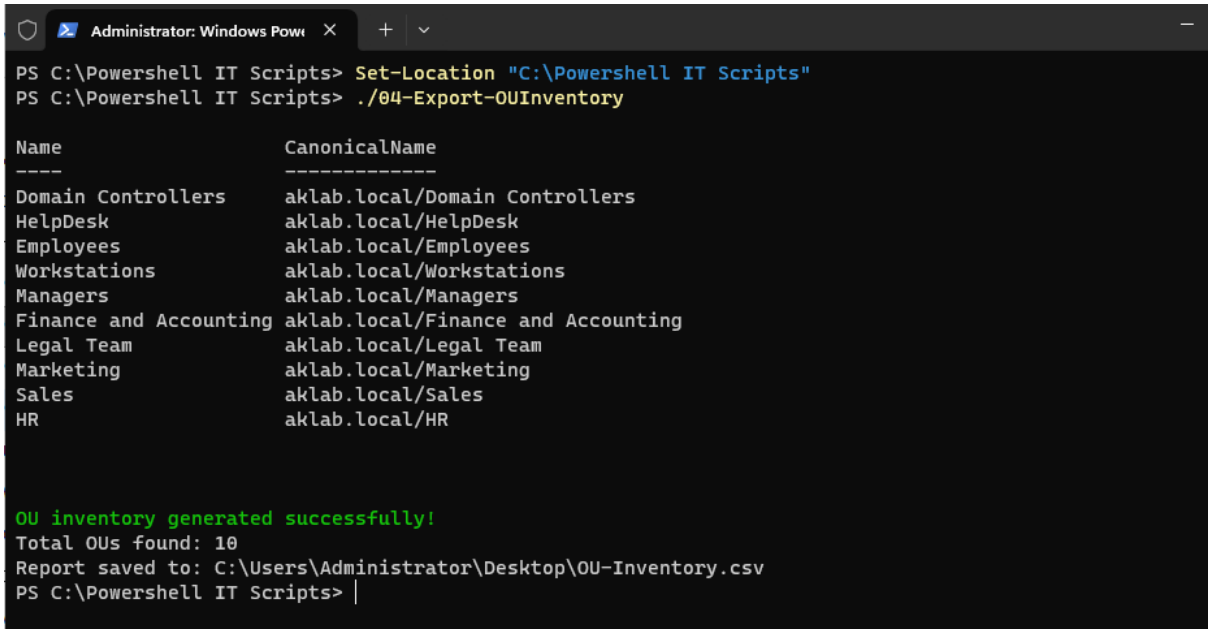


Figure 13: Script 4 runs successfully

Running Script 5:

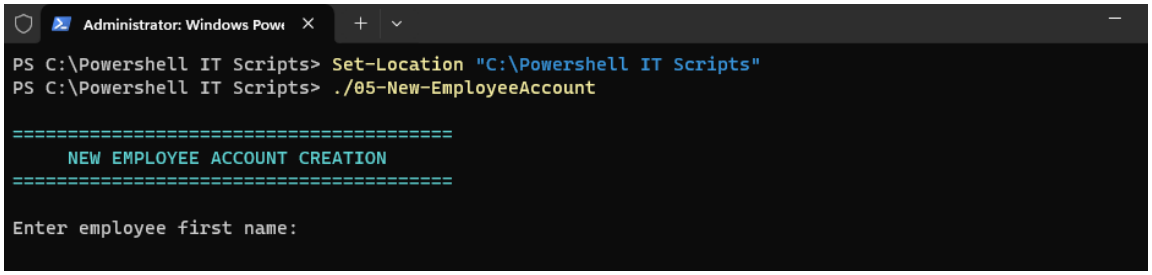


Figure 14: Script 5 runs successfully for the first part

Upon running, it asks for the details one by one and at the end ask for confirmation to create the account with the details just mentioned.

```
=======================================
     NEW EMPLOYEE ACCOUNT CREATION
=======================================

Enter employee first name: Alicia
Enter employee last name: Heatherson
What username do you want?: alicia.heatherson
Enter employee department: Employees

Available Organizational Units:

Name                   DistinguishedName
----                   -----------------
Domain Controllers     OU=Domain Controllers,DC=aklab,DC=local
HelpDesk               OU=HelpDesk,DC=aklab,DC=local
Employees              OU=Employees,DC=aklab,DC=local
Workstations           OU=Workstations,DC=aklab,DC=local
Managers               OU=Managers,DC=aklab,DC=local
Finance and Accounting OU=Finance and Accounting,DC=aklab,DC=local
Legal Team             OU=Legal Team,DC=aklab,DC=local
Marketing              OU=Marketing,DC=aklab,DC=local
Sales                  OU=Sales,DC=aklab,DC=local
HR                     OU=HR,DC=aklab,DC=local


Enter the target OU Distinguished Name: OU=Employees,DC=aklab,DC=local
Enter initial password: ********************

========== REVIEW DETAILS ==========
Full name:  Alicia Heatherson
Username:   alicia.heatherson
Department: Employees
Target OU:  OU=Employees,DC=aklab,DC=local
====================================
Type YES to create this employee account:
```

Figure 15: Account confirmation

After reviewing the details, confirming YES, the account is created:

```
========== REVIEW DETAILS ==========
Full name:  Alicia Heatherson
Username:   alicia.heatherson
Department: Employees
Target OU:  OU=Employees,DC=aklab,DC=local
====================================
Type YES to create this employee account: YES

Employee account created successfully!
Username: alicia.heatherson
Domain: aklab.local
PS C:\Powershell IT Scripts>
```

Figure 16: The account is created

Let’s verify in the directory if Alicia Heatherson’s name exists there:

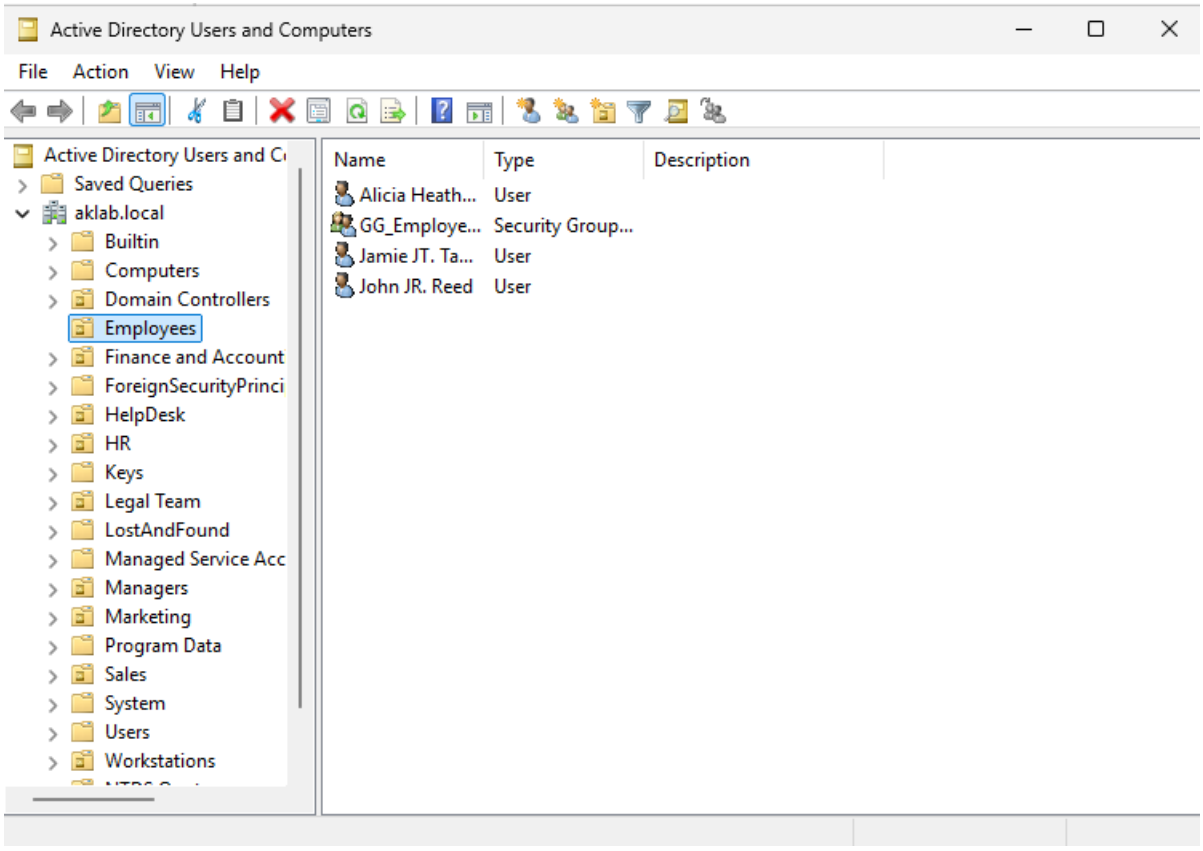


Figure 17: Employee Account verified

Alicia Heatherson's account exists in the Employees folder.

Hence, all the scripts are working.

Ticket Resolved!